Technology stack.

Frontend — React + Vite

Builds the pages, department cards, search, question viewer, upload forms, and admin dashboard.

Backend — Node.js + Express.js

Handles APIs, uploads, duplicate checks, publication rules, authentication checks, and admin permissions.

Database — PostgreSQL

Stores departments, university sessions, semesters, courses, question records, users, permissions, and audit history.

File storage — Supabase Storage

Stores PDF, image, and document files separately from the database. Pending uploads remain private.

UI styling — Tailwind CSS + custom CSS

Creates responsive glassmorphism cards, gradients, blur effects, animations, and mobile layouts.